MALLA REDDY UNIVERSITY

AI PREVENTIVE MAINTENANCE SYSTEM

Real-time Monitoring & Predictive Failure Detection

Guided by: Mrs. B. Swetha

Department of Electronics & Communication Engineering | Malla Reddy University, Hyderabad

Introduction

The motivation for this project is to prevent machine failures and reduce maintenance costs through an IoT-enabled predictive maintenance system. Predictive maintenance uses sensor data to identify patterns and predict possible failures before they occur.

- Predicting system issues before they occur.
- Predicting system maintenance needs before failure.
- Providing background information on the project.

Objectives

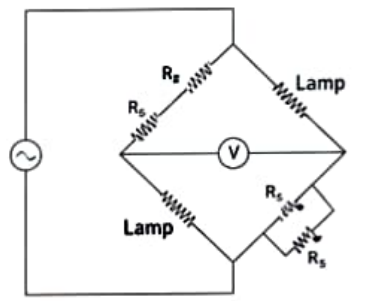
- Minimize downtime and maintenance costs.
- Create low-cost predictive hardware using sensors.

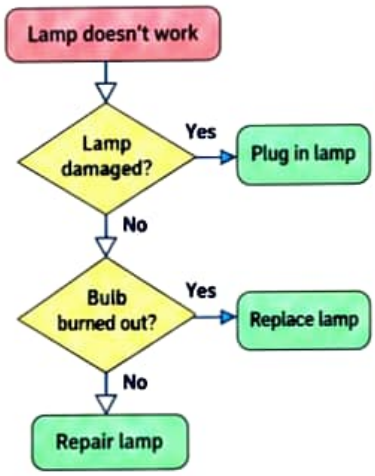
Societal Impact

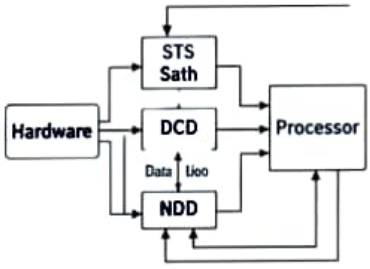
- Early fault detection can help prevent accidents and reduce industrial waste by identifying faults before they cause serious problems.

Methodology

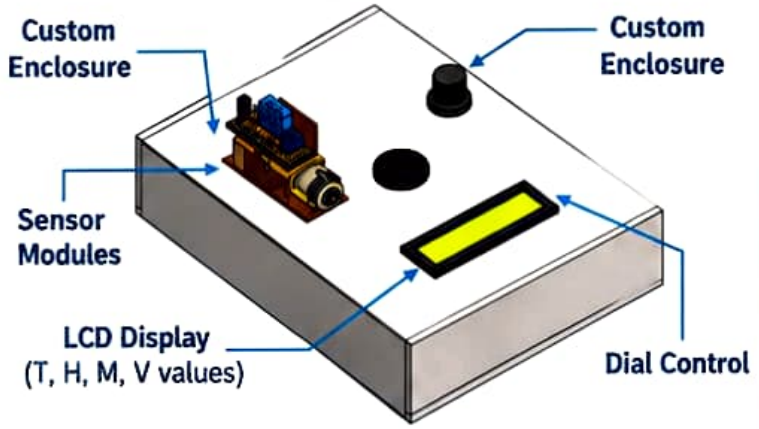
Real-time data collection using hardware sensors and embedded systems. The sensors monitor machine conditions and collect data for analysis. This section may be supported by circuit diagrams and flowcharts. Multiple figures may be inserted to show the process.







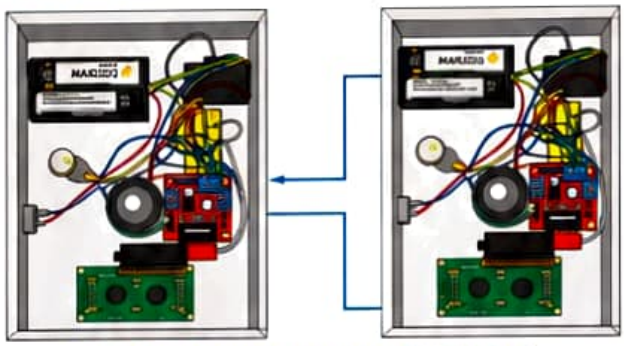
Flowchart to hardware hardware to schematic



Student Prototype

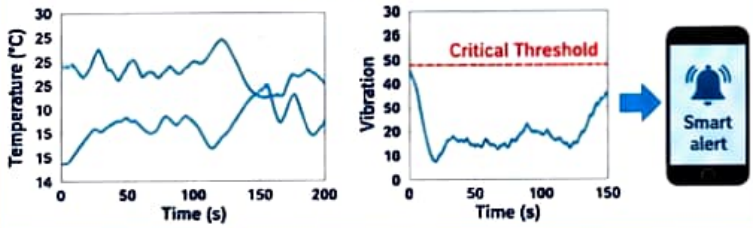
Experimental Setup & Outcomes

- Calibration and sensor measurements were performed using the prototype.
- Readings were monitored and recorded using the microcontroller.



Internal Hardware Configuration

Digital Dashboard



Conclusions & Future Work



Real-time Alerts & Notification Screen

Course: B.Tech Major Project
Department: Electronics & Communication Engineering
Year: 2026–2027